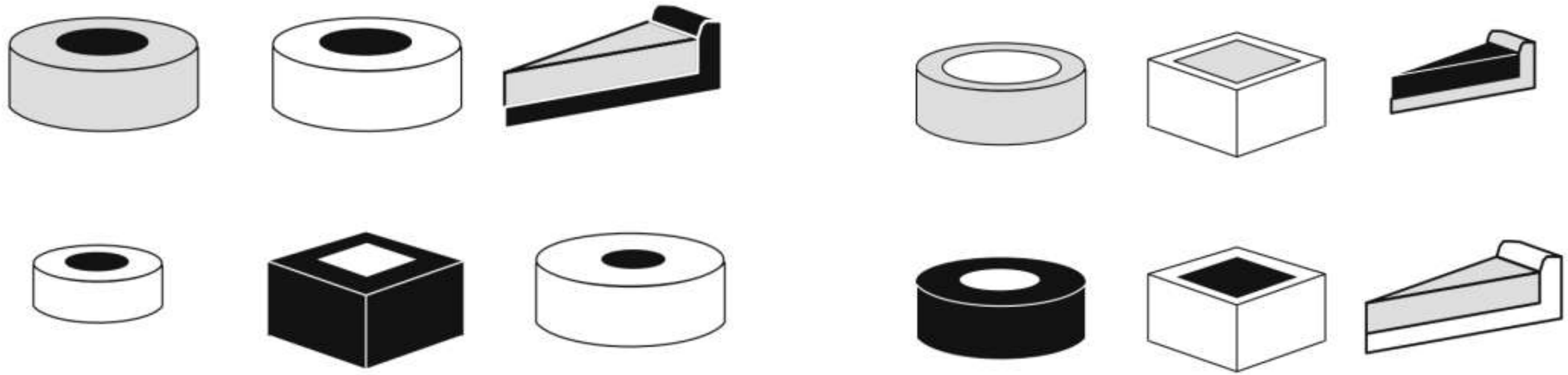
Training set, Classes and Classifier:



Training set: Positive and negative examples: Sunil likes or does NOT like.

Classes: Positive and negative

Classifier: An algorithm for Categorizing (future pie into Classes):

Dataset:

Atrribute-Value = 5

Data Set = 12

Classes = 2

Example	Shape	Crust		Filling		Class
		Size	Shade	Size	Shade	
ex1	Circle	Thick	Gray	Thick	Dark	pos
ex2	Circle	Thick	White	Thick	Dark	pos
ex3	Triangle	Thick	Dark	Thick	Gray	pos
ex4	Circle	Thin	White	Thin	Dark	pos
ex5	Square	Thick	Dark	Thin	White	pos
ex6	Circle	Thick	White	Thin	Dark	pos
ex7	Circle	Thick	Gray	Thick	White	neg
ex8	Square	Thick	White	Thick	Gray	neg
ex9	Triangle	Thin	Gray	Thin	Dark	neg
ex10	Circle	Thick	Dark	Thick	White	neg
ex11	Square	Thick	White	Thick	Dark	neg
ex12	Triangle	Thick	White	Thick	Gray	neg

Shape (circle, triangle, and square)

Crust-size (thin or thick),

Crust-shade (white, gray, or dark),

Filling-size (thin or thick), and

Filling-shade (white, gray, or dark)

(shape=circle) AND (crust-size=thick) AND (crust-shade=gray) AND (filling-size=thick) AND (filling-shade=dark)

Dataset: Training, Validation and Testing Datasets

Initial dataset is split into two parts:

1. A training set
2. A testing set

Training dataset:

The model sees and learns from this data to predict the outcome or to make the right decisions

The model can also correct itself when predictions are wrong

Testing dataset:

Only used for evaluating your algorithm.

the training set and testing set are *independent of each other* **and *do not overlap!***

Neural networks have a number parameters (ex., learning rate, decay, regularization, etc.)

In practice, we need to test a bunch of these hyperparameters and identify the set of parameters that works the best.

create a *third* data split called the validation set

We normally allocate roughly 10-20% of the training data for validation

Dipanjan Sarkar, Raghav Bali, Tushar Sharma

Training and Testing sets:

Training Set = 8

TestingSet = 4

Example	Shape	Crust Size	Crust Shade	Filling Size	Filling Shade	Class
ex1	Circle	Thick	Gray	Thick	Dark	pos
ex2	Circle	Thick	White	Thick	Dark	pos
ex3	Triangle	Thick	Dark	Thick	Gray	pos
ex4	Circle	Thin	White	Thin	Dark	pos
ex5	Square	Thick	Dark	Thin	White	pos
ex6	Circle	Thick	White	Thin	Dark	pos
ex7	Circle	Thick	Gray	Thick	White	neg
ex8	Square	Thick	White	Thick	Gray	neg
ex9	Triangle	Thin	Gray	Thin	Dark	neg ✓
ex10	Circle	Thick	Dark	Thick	White	neg ✗
ex11	Square	Thick	White	Thick	Dark	neg ✓
ex12	Triangle	Thick	White	Thick	Gray	neg ✓

The Classifier performance is estimated as 75%

M. Kubat, "Introduction to ML"

different examples

Instance space:

Example	Shape	Crust		Filling		Class
		Size	Shade	Size	Shade	
ex1	Circle	Thick	Gray	Thick	Dark	pos
ex2	Circle	Thick	White	Thick	Dark	pos
ex3	Triangle	Thick	Dark	Thick	Gray	pos
ex4	Circle	Thin	White	Thin	Dark	pos
ex5	Square	Thick	Dark	Thin	White	pos
ex6	Circle	Thick	White	Thin	Dark	pos
ex7	Circle	Thick	Gray	Thick	White	neg
ex8	Square	Thick	White	Thick	Gray	neg
ex9	Triangle	Thin	Gray	Thin	Dark	neg
ex10	Circle	Thick	Dark	Thick	White	neg
ex11	Square	Thick	White	Thick	Dark	neg
ex12	Triangle	Thick	White	Thick	Gray	neg

$3 \times 2 \times 3 \times 2 \times 3 = 108$ different examples

Each example subsets may represent list of positive examples of someone's notion of a "good pie."

M. Kubat, "Introduction to ML"

Classifier:

Example	Shape	Crust		Filling		Class
		Size	Shade	Size	Shade	
ex1	Circle	Thick	Gray	Thick	Dark	pos
ex2	Circle	Thick	White	Thick	Dark	pos
ex3	Triangle	Thick	Dark	Thick	Gray	pos
ex4	Circle	Thin	White	Thin	Dark	pos
ex5	Square	Thick	Dark	Thin	White	pos
ex6	Circle	Thick	White	Thin	Dark	pos
ex7	Circle	Thick	Gray	Thick	White	neg
ex8	Square	Thick	White	Thick	Gray	neg
ex9	Triangle	Thin	Gray	Thin	Dark	neg
ex10	Circle	Thick	Dark	Thick	White	neg
ex11	Square	Thick	White	Thick	Dark	neg
ex12	Triangle	Thick	White	Thick	Gray	neg

[(shape=circle) AND (filling-shade=dark)]

Classifier: [(shape=circle) AND (filling-shade=dark)] OR
[NOT(shape=circle) AND (crust-shade=dark)]

M. Kubat, “Introduction to ML”

Validation / Development Dataset

Data set derived for Training dataset

- Assess model's performance and tune hyperparameters
- Unbiased evaluation of fully tuned model (intermediate state)
- It is used to avoid overfitting by monitoring the training performance
- Common practice approximately 10%-20% of training dataset but depends on model hyperparameters.

different examples

Instance space:

Example	Shape	Crust		Filling		Class
		Size	Shade	Size	Shade	
ex1	Circle	Thick	Gray	Thick	Dark	pos
ex2	Circle	Thick	White	Thick	Dark	pos
ex3	Triangle	Thick	Dark	Thick	Gray	pos
ex4	Circle	Thin	White	Thin	Dark	pos
ex5	Square	Thick	Dark	Thin	White	pos
ex6	Circle	Thick	White	Thin	Dark	pos
ex7	Circle	Thick	Gray	Thick	White	neg
ex8	Square	Thick	White	Thick	Gray	neg
ex9	Triangle	Thin	Gray	Thin	Dark	neg
ex10	Circle	Thick	Dark	Thick	White	neg
ex11	Square	Thick	White	Thick	Dark	neg
ex12	Triangle	Thick	White	Thick	Gray	neg

$3 \times 2 \times 3 \times 2 \times 3 = 108$ different examples

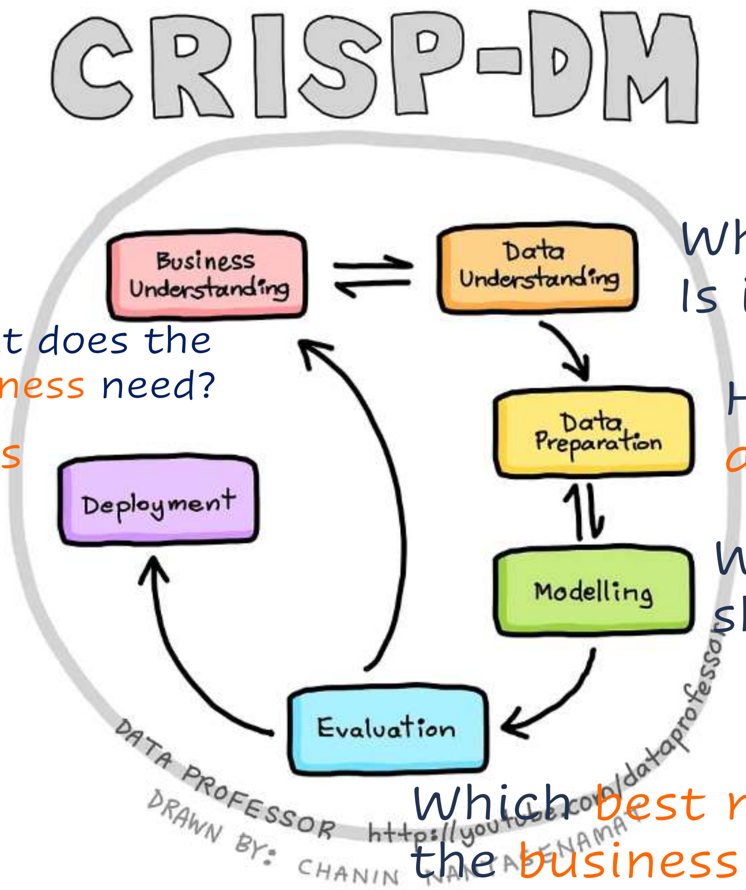
Each example subsets may represent list of positive examples of someone's notion of a "good pie."

M. Kubat, "Introduction to ML"

CRISP-DM framework: CRoss Industry Standard Process for Data Mining

CRISP-DM was conceived in 1996 and became a European Union project under the ESPRIT funding initiative in 1997.

The project was led by five companies: Integral Solutions Ltd (ISL), Teradata, Daimler AG, NCR Corporation and OHRA, an insurance company.



What does the business need?

What data do we have / need? Is it clean?

How do we organize the data for modeling?

What modeling techniques should we apply?

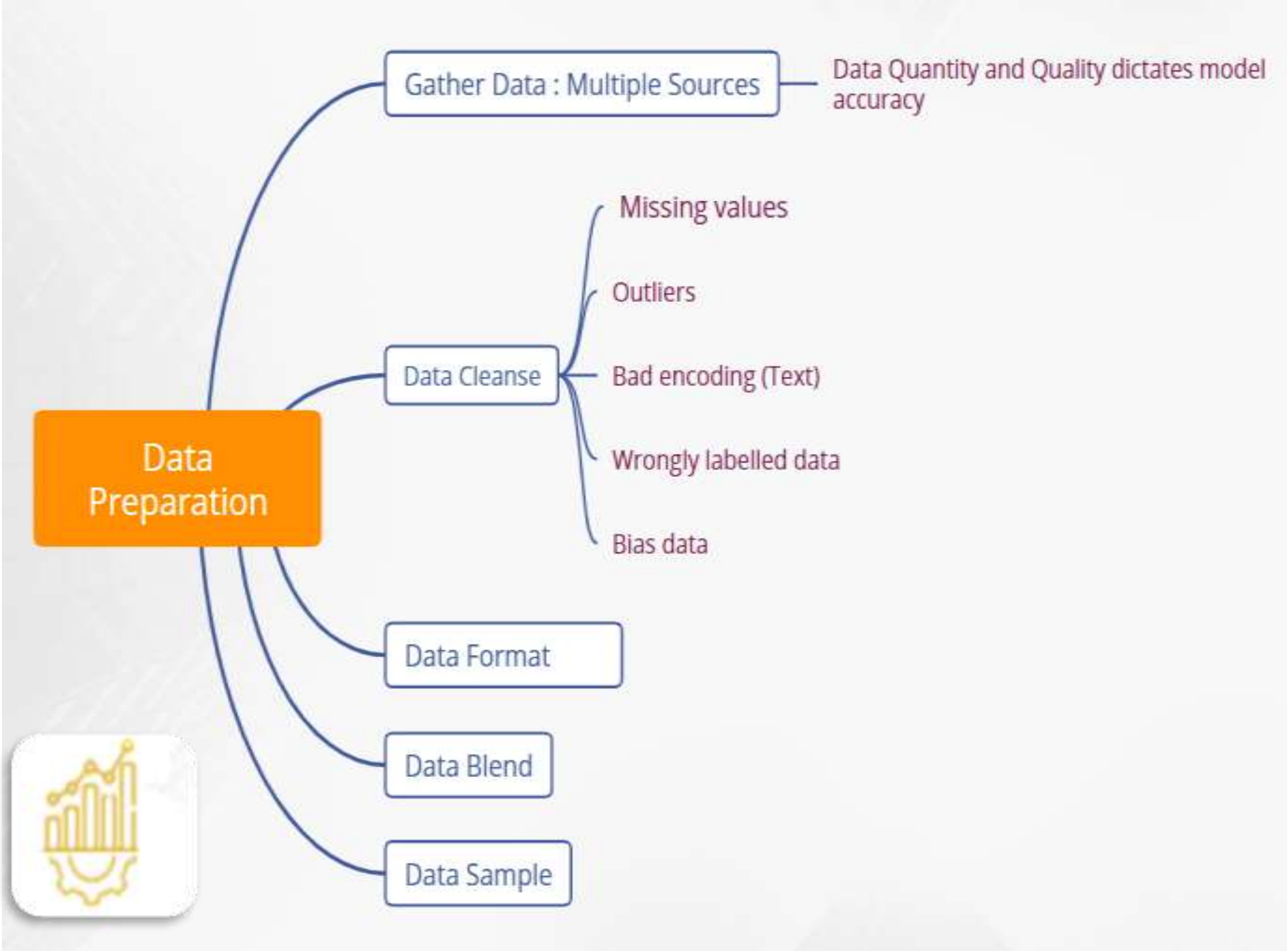
Which best model meets the business objectives?

How do stakeholders access the results?

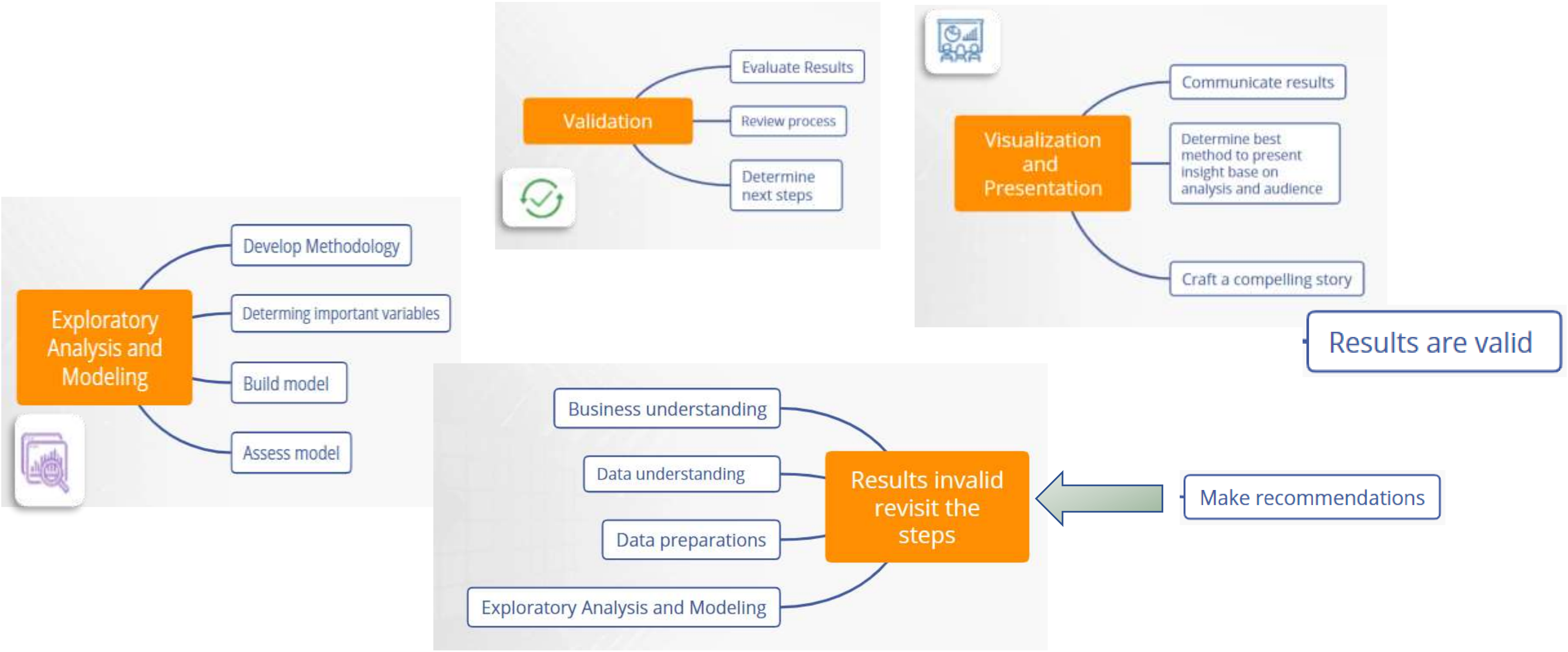
It is a process model that serves as the base for a data science process.

It has six sequential phases:

Data analysis project life cycle:



Data analysis project life cycle:



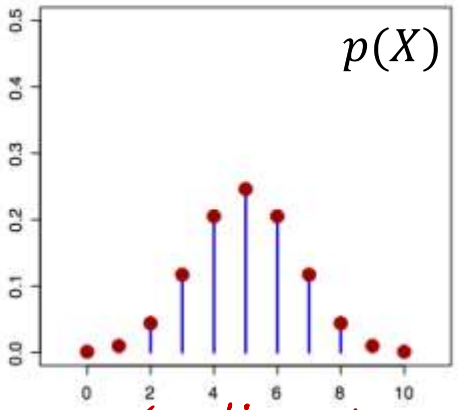
Random variables:

- A random variable (r.v.) X denotes possible outcomes of an event

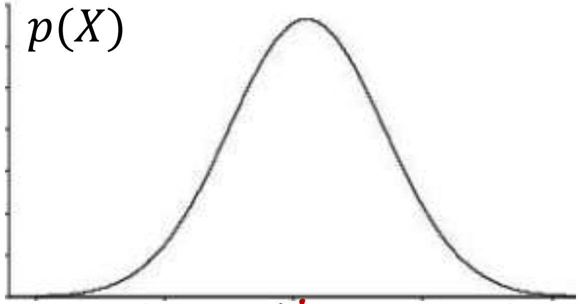
- It can be discrete (i.e., finite many possible outcomes) or continuous

- Some examples of discrete r.v.
  - $X \in \{0,1\}$ denoting outcomes of a coin-toss
  - $X \in \{1,2,\ldots,6\}$ denoting outcome of a dice roll

- Some examples of continuous r.v.
  - $X \in (0,1)$ denoting the bias of a coin
  - $X \in \mathbb{R}$ denoting heights of students in the class



X(a discrete r.v.)



X(a continuous r.v.)

Discrete Random Variable:

- For a discrete r.v. X, $p(x)$ denotes $p(X = x)$

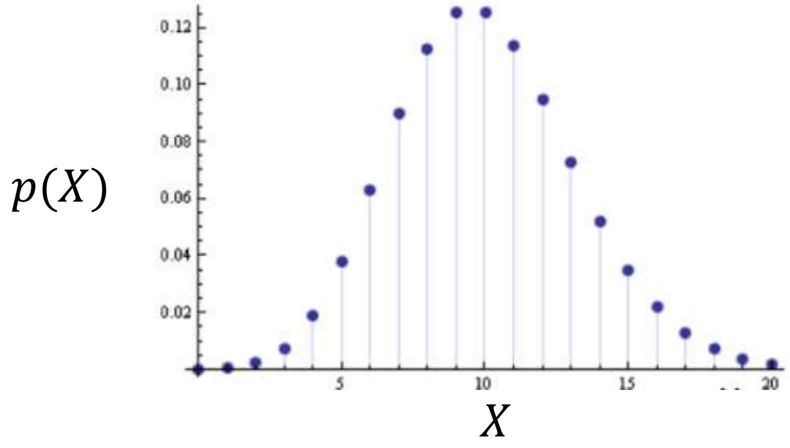
    probability that $X = x$

- $p(X)$ is called the probability mass function (PMF) of r.v. X
    - $p(x)$ or $p(X = x)$ is the value of the PMF at x

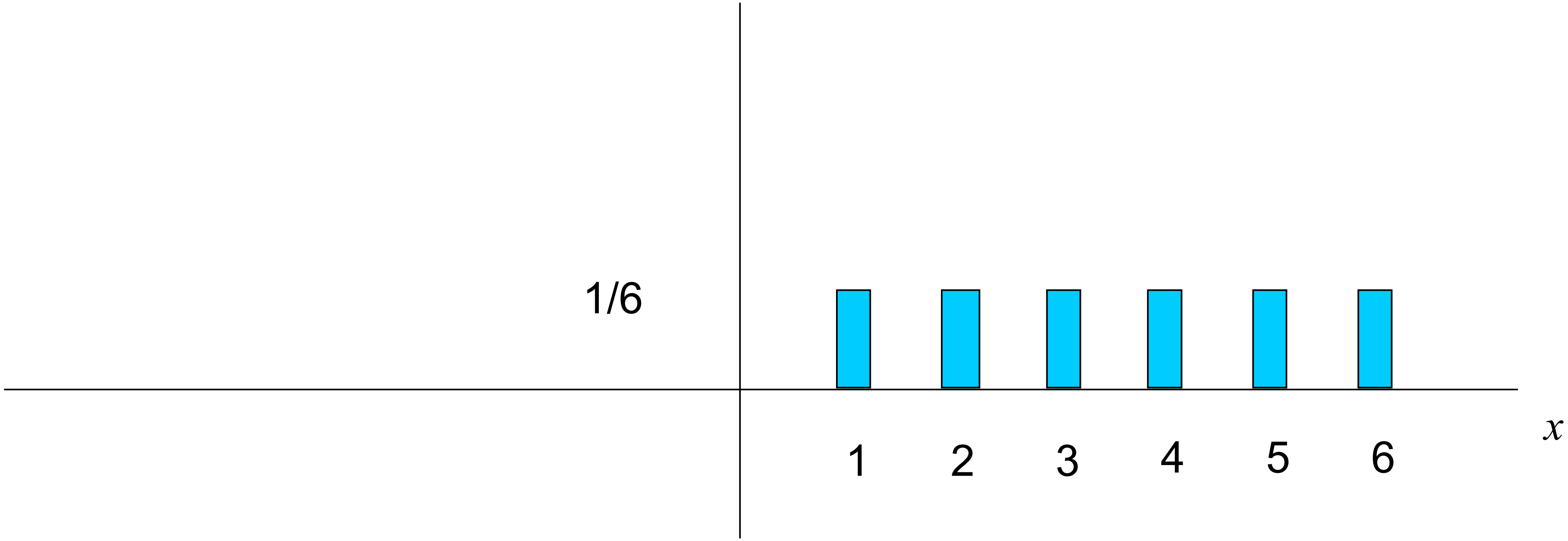
$$p(x) \geq 0$$

$$p(x) \leq 1$$

$$\sum_x p(x) = 1$$



Discrete example: roll of a dice



$$\sum_{\text{all } x} P(x) = 1$$

Probability mass function (Pmf):

x	*p(x)*
1	*p(x=1)*=1/6
2	*p(x=2)*=1/6
3	*p(x=3)*=1/6
4	*p(x=4)*=1/6
5	*p(x=5)*=1/6
6	*p(x=6)*=1/6

Joint and Conditional Probabilities:

Joint probability of two events occurring simultaneously.

$P(\text{pos}, \text{thick})$ the probability that the example is positive **and** its filling is thick

conditional probability of two events

$P(\text{pos}|\text{thick})$ refers to the occurrence of a positive example among those that have filling-size=thick.

Prior Probability:

$P(\text{pos}) = 6/12$ is obtained by dividing the pos class (six) with that of the entire training set (twelve)

Probabilities: Pie example

Example	Shape	Crust		Filling		Class
		Size	Shade	Size	Shade	
ex1	Circle	Thick	Gray	Thick	Dark	pos
ex2	Circle	Thick	White	Thick	Dark	pos
ex3	Triangle	Thick	Dark	Thick	Gray	pos
ex4	Circle	Thin	White	Thin	Dark	pos
ex5	Square	Thick	Dark	Thin	White	pos
ex6	Circle	Thick	White	Thin	Dark	pos
ex7	Circle	Thick	Gray	Thick	White	neg
ex8	Square	Thick	White	Thick	Gray	neg
ex9	Triangle	Thin	Gray	Thin	Dark	neg
ex10	Circle	Thick	Dark	Thick	White	neg
ex11	Square	Thick	White	Thick	Dark	neg
ex12	Triangle	Thick	White	Thick	Gray	neg

$(N_{thick} = 8)$

The training set consists of twelve pies (N=1

six are positive examp $(N_{pos} = 6)$

six are negative $(N_{neg} = 6)$

The prior probability of Johnny liking a randomly picked pie is

$$P(\text{pos}) = \frac{N_{pos}}{N_{all}} = \frac{6}{12} = 0.5$$

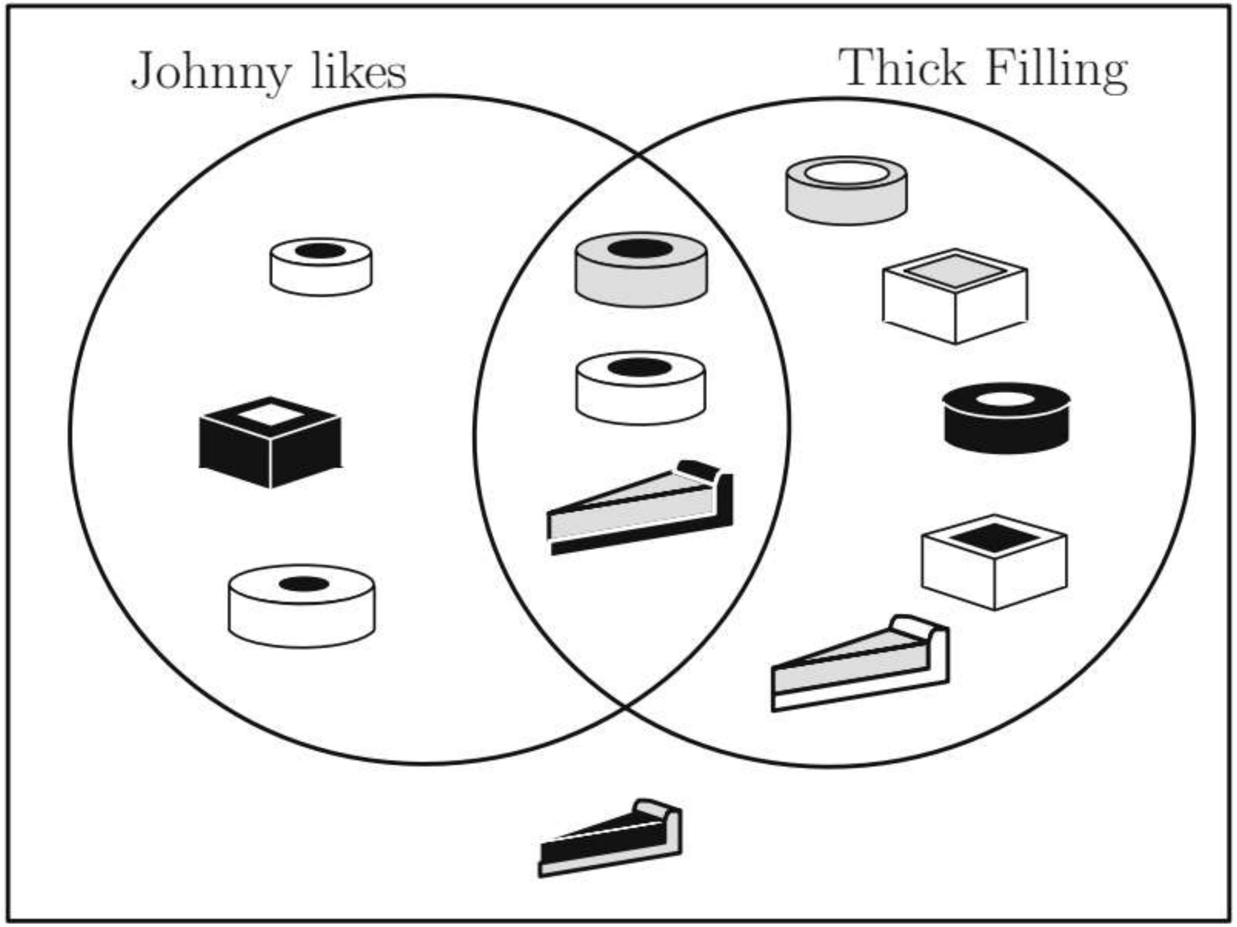
Out of these, three are labeled as $(N_{pos|thick} = 3)$ positive

$$P(\text{pos}|\text{thick}) = \frac{N_{pos|thick}}{N_{thick}} = \frac{3}{8} = 0.375$$

"*conditional probability* of an example being positive given that filling-size=thick" is 37.5%

M. Kubat, "Introduction to ML"

Probabilities: Pie example



$$P(\texttt{pos}) = \frac{6}{12}$$

$$P(\texttt{thick}) = \frac{8}{12}$$

$$P(\texttt{pos}|\texttt{thick}) = \frac{3}{8}$$

$$P(\texttt{thick}|\texttt{pos}) = \frac{3}{6}$$

joint probability,
$$P(\texttt{likes}, \texttt{thick}) = \frac{3}{12}$$

M. Kubat, “Introduction to ML”

Quiz: Prior and Conditional Probabilities

You are given 8 examples with feature "Size" and Class (Table given below)

	ex1	ex2	ex3	ex4	ex5	ex6	ex7	ex8
Size	thick	thick	thin	thin	thin	thick	thick	thick
Class	pos	pos	pos	pos	neg	neg	neg	neg

Write a Python code for computing the prior and conditional probabilities given below:

$P(\text{thin}) =$

$P(\text{thick})$

$P(\text{pos}) =$

$P(\text{neg}) =$

$P(\text{thin}|\text{pos}) =$

$P(\text{thick}|\text{pos})$

$P(\text{thin}|\text{neg}) =$

$P(\text{thick}|\text{neg})$

Conditional Probability

The conditional probability of a concrete attribute value within a given class is, again, determined by relative frequency

$P(\text{pos}|\text{chick})$.

Applying conditional probabilities to Classifier

Example	Shape	Crust		Filling		Class
		Size	Shade	Size	Shade	
ex1	Circle	Thick	Gray	Thick	Dark	pos
ex2	Circle	Thick	White	Thick	Dark	pos
ex3	Triangle	Thick	Dark	Thick	Gray	pos
ex4	Circle	Thin	White	Thin	Dark	pos
ex5	Square	Thick	Dark	Thin	White	pos
ex6	Circle	Thick	White	Thin	Dark	pos
ex7	Circle	Thick	Gray	Thick	White	neg
ex8	Square	Thick	White	Thick	Gray	neg
ex9	Triangle	Thin	Gray	Thin	Dark	neg
ex10	Circle	Thick	Dark	Thick	White	neg
ex11	Square	Thick	White	Thick	Dark	neg
ex12	Triangle	Thick	White	Thick	Gray	neg

conditional probability, $P(\text{pos}|\text{thick})$, is more trustworthy than the prior probability, $P(\text{pos})$.

$P(\text{pos}|\text{thick}) = 3/8 = 0.375$

$P(\text{neg}|\text{thick}) = 5/8 = 0.625$

$$\boxed{P(\text{neg}|\text{thick}) > P(\text{pos}|\text{thick})}$$

Conclusíon: The probability of Johnny disliking a pie with thick filling is greater

The classifier to label all examples with filling-size=thick as negative instances of the "pie that Johnny likes."

M. Kubat, "Introduction to ML"

Joint probabilities from Prior and Conditional probabilities

$$P(\texttt{pos}, \texttt{thick}) = P(\texttt{pos}|\texttt{thick}) \cdot P(\texttt{thick}) = \frac{3}{8} \cdot \frac{8}{12} = \frac{3}{12}$$

$$P(\texttt{thick}, \texttt{pos}) = P(\texttt{thick}|\texttt{pos}) \cdot P(\texttt{pos}) = \frac{3}{6} \cdot \frac{6}{12} = \frac{3}{12}$$

Joint probability can NEVER EXCEED the value of the corresponding **Conditional probability**

$$P(\texttt{pos}, \texttt{thick}) \leq P(\texttt{pos}|\texttt{thick})$$

Conditional Probability is multiplied by prior probability

$P(\texttt{thick})$ or $P(\texttt{pos})$

which can never be greater than 1

M. Kubat, "Introduction to ML"

Facts:

$$P(\texttt{thick}, \texttt{pos}) \;=\; P(\texttt{pos}, \texttt{thick})$$

both represent the same thing:

the probability of **thick** and **pos** co-occurring

$$P(\texttt{pos}, \texttt{thick}) = P(\texttt{pos}|\texttt{thick}) \cdot P(\texttt{thick}) = \frac{3}{8} \cdot \frac{8}{12} = \frac{3}{12}$$

$$P(\texttt{thick}, \texttt{pos}) = P(\texttt{thick}|\texttt{pos}) \cdot P(\texttt{pos}) = \frac{3}{6} \cdot \frac{6}{12} = \frac{3}{12}$$

Consequently, the left-hand sides of the previous two formulas have to be equal, which implies the following:

$$P(\texttt{pos}|\texttt{thick}) \cdot P(\texttt{thick}) = P(\texttt{thick}|\texttt{pos}) \cdot P(\texttt{pos})$$

Dividing both sides

$$P(\texttt{pos}|\texttt{thick}) = \frac{P(\texttt{thick}|\texttt{pos}) \cdot P(\texttt{pos})}{P(\texttt{thick})}$$

Bayes Formula

A hypothesis is an assumption, an idea that is proposed for the sake of argument so that it can be tested to see if it might be true

Bayes rule:

Daily exposure to the sun leads to increased levels of happiness.

Evidence is the information to establish or refute claims relevant to a case

It is the mathematical rule that describes how to update a belief, given some evidence

$$\underbrace{\boxed{P(H \mid E)}}_{\text{Posterior}} = \underbrace{\boxed{\frac{P(E \mid H)}{P(E)}}}_{\text{likelihood ratio}} \underbrace{\boxed{P(H)}}_{\text{Prior}}.$$

Posterior probability (updated probability of hypothesis (H) after the evidence is considered)

Prior probability (the probability of hypothesis (H) before the evidence (E) is considered)

Likelihood (probability of the evidence (E), given the belief/hypothesis is true)
Marginal probability (probability of the evidence (E) , under any circumstance)

$$\frac{\overbrace{\boxed{P(E \mid H)}}^{\text{likehood}}}{\underbrace{\boxed{P(E)}}_{\text{marginal}}}$$

spam filters use Bayesian updating to determine whether an email is real or spam, given the words in the email.

Bayes' Rule applications:

- ✓ Understanding probability problems in medical research
- ✓ Statistical modelling and inference (Inference applications: philosophy, sport, science and law).
- ✓ Machine learning algorithms (such as Naive Bayes, Expectation Maximization)
- ✓ Quantitative modelling and forecasting

Naives Bayers Classifier (NBC): Solution

Day	*Outlook*	*Temperature*	*Humidity*	*Wind*	*PlayTennis*
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Prior pobablities (Pp)

Pp(PlayTennis = yes) = 9/14 = 0.64

Pp(PlayTennis = no) = 5/14 = 0.36

Outlook	Yes	No
sunny	2/9	3/5
overcast	4/9	0
rain	3/9	2//5

Temperature	Yes	No
hot	2/9	2/5
mild	4/9	2/5
cool	3/9	1/5

Humidity	Yes	No
high	3/9	4/5
normal	6/9	1/5

Windy	Yes	No
Stron	3/9	3/5
Weak	6/9	2/5

Conditional pobablities

Test Data: Solution

[Outlook = sunny, Temperature = cool, Humidity = high, Wind = strong]

Tnb (yes) = Pp(play Tennis = yes) P(sunny| yes) P(cool|yes) P(high|yes) P(strong| yes) = 0.0053

Tnb (no) = Pp(Play Tennis = no) P(sunny| no) P(cool|no) P(high|no) P(strong| no) = 0.0206

Normalized Propability

Tnb (yes) = Tnb (yes) / (Tnb (yes) + Tnb (no)) = 0.205

Tnb (no) = Tnb (no) / (Tnb (yes) + Tnb (no)) = 0.795

The test instance is classified as NOT to play tennis

TECHNISCHE HOCHSCHULE DEGGENDORF THD Campus Cham

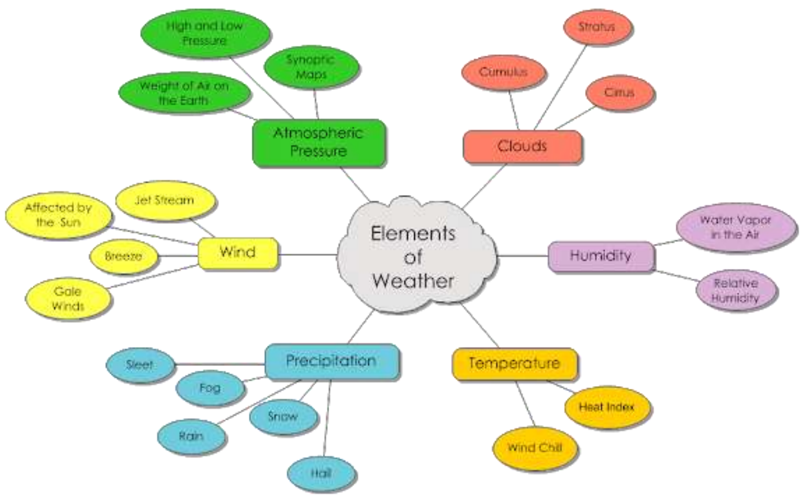
Naives Bayers Classifier (NBC): Applications

LiDAR-based Obstacle Detection:

To distinguish between obstacles and open space, allowing mobile robots to navigate complex environments.

Sensor Fusion (Integrating data from multiple sensors (e.g., cameras, sonar, LiDAR),

Naive Bayes can provide a probabilistic assessment of the environment, aiding in semantic mapping.



Speed and Efficiency:

Fast training and inference times allow for real-time operation.

Robustness:

They handle noisy and incomplete sensor data effectively.

Low Memory Footprint:

They are suitable for embedded systems with limited resources.